

Rowan Yu

Senior Unreal Engine Software Engineer | XR Systems | Developer Tooling | Performance

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PROFESSIONAL SUMMARY

Senior Unreal Engine Software Engineer with 8+ years of experience building VR/XR simulation systems, multiplayer architectures, developer-facing OpenXR plugins/samples, reusable Unreal tools and performance-sensitive real-time applications. At Big X Reality, developed military VR training simulators for mixed PC and PC-to-VR streamed workflows supporting up to 30 connected computers. Former HTC VIVE R&D engineer with hands-on experience creating ViveOpenXR Unreal plugins, developer samples and technical tutorials. Recently validated an Unreal Android build/deploy workflow by packaging an APK, deploying it to VIVE Focus 3 via ADB and inspecting runtime logs with Logcat.

CORE SKILLS

Unreal Engine 4/5, C++, Blueprint, C#, Unity | SDK/plugin architecture: ViveOpenXR, SRWorks Unreal Plugin, reusable Unreal plugins, developer samples, technical tutorials | Android validation: Unreal Android packaging, APK deployment to VIVE Focus 3, ADB, Logcat, Android SDK/NDK workflow | Multiplayer: Replication, RPC, server/client launch workflows, cross-client synchronization, PC-to-VR streamed workflows | XR/VR: OpenXR, SteamVR, VIVE Pro 2, VIVE Focus 3, VIVE Tracker, SRWorks | Performance: Unreal Insights, profiling, async processing, multithreading, NPC optimization, runtime tuning | Tools: packaged-build launch tools, external config/Excel tuning, data collection workflows, QA/test utilities | Hardware/data integration: shared-memory IO, motion platform, Arduino, Antilatency, OptiTrack, HTTP, Socket.IO, TCP/UDP | Collaboration: mentoring, ad-hoc code/Blueprint review, cross-functional debugging, developer/customer validation support | Languages: Mandarin Chinese (Native), English (Professional working proficiency; experience collaborating with international colleagues)

PROFESSIONAL EXPERIENCE

Big X Reality - Senior Unreal Engineer | 2022/08 - Present | New Taipei City, Taiwan

- Developed multiplayer VR training simulators for high-fidelity Windows PC and PC-to-VR streamed workflows, including VIVE Focus 3 environments, with performance-sensitive rendering and client/server synchronization.
- Developed large-scale multiplayer training systems supporting up to 30 connected computers across desktop and PC-VR clients; used Unreal Insights to trace and resolve performance and network transmission bottlenecks.
- Validated Unreal Android packaging by building an APK, deploying it to VIVE Focus 3 through ADB and inspecting runtime behavior through Logcat as a focused Android ramp-up exercise.
- Owned NPC behavior design and implementation for training scenarios, and designed reusable Unreal plugins/components for NPC performance optimization, data collection and project-wide content generation workflows.
- Created a data collection abstraction layer that allowed teammates to attach telemetry capture to gameplay features, then transformed, encrypted, timestamped and routed data through HTTP/Socket.IO pipelines to backend systems.
- Co-developed Antilatency and motion-platform integration plugins, including shared-memory signal exchange for real-time IO and platform control to support more realistic vehicle and speedboat simulation feedback.
- Built packaged-build testing tools such as .bat launchers for dedicated servers and selected client types, allowing teammates to quickly reproduce multiplayer scenarios without manually launching each executable.
- Built runtime configuration tools that allowed non-Unreal teammates to tune packaged EXE experiences through external Excel/config files during customer validation sessions, avoiding rebuilds and enabling faster feedback cycles.
- Mentored 2 junior engineers on Unreal development practices, Perforce, Jenkins, Git setup and project workflows; provided ad-hoc C++ and Blueprint reviews when teammates needed technical validation.

HTC Corporation - Engineer, VIVE R&D | 2019/06 - 2022/08 | New Taipei City, Taiwan

- Developed SRWorks UE4 Experience, a VR sample project built for HTC SRWorks Runtime to help Unreal developers quickly understand and use SRWorks features in their own XR projects.
- Ported SRWorks Scene Understanding C++ low-level APIs into the OpenXR architecture, adapting API behavior and integration patterns to comply with OpenXR design rules.
- Developed ViveOpenXR Plugin for UE4.27 and UE5.0, including OpenXR Scene Understanding, OpenXR Facial Tracking and VIVE Cosmos controller mapping support.
- Implemented the OpenXR Facial Tracking Module using Unreal Live Link to stream facial expression data into Unreal Engine animation workflows.
- Created developer-facing ViveOpenXR Samples for OpenXR Facial Tracking, Avatar/MetaHuman integration, Scene Understanding, Hand Tracking and Eye Gaze workflows.
- Authored Unreal-focused ViveOpenXR Tutorials covering Facial Tracking, MetaHuman integration, Scene Understanding and Hand Tracking, reducing developer adoption friction for HTC-supported OpenXR features.

Artlord Studio - Engineer, R&D | 2016/07 - 2018/12 | Kaohsiung, Taiwan

- Built multiplayer VR shooting game systems using UE4 Replication, VR IK and profiling-based network traffic reduction for commercial VIVELAND deployment.
- Integrated HTC VIVE with Kinect, Fingo hand tracking and a single-axis motion platform across multiple location-based entertainment projects.

COMMUNITY & GAME DEVELOPMENT

- Expanded a Global Game Jam prototype into a location-based VR entertainment experience integrated with a single-axis motion platform.

EDUCATION

Southern Taiwan University of Science and Technology - B.S. Multimedia and Entertainment Science